MOBILE COMPUTING APPLICATIONS

Internet has been labeled as the path breaking technology. It has brought a dramatic change in business landscape and touched the life of common man.

Mobile computing and wireless communication networks are playing an increasingly important role in our professional and personal lives. Nowadays we use many mobile devices like mobiles, PDAs etc. for personal and business communications.

Applications of mobile computing are almost limitless and exist in socio-economic, industrial and commercial domains.

How various mobile computing applications are playing a key role in improving our life.

1. In business

M – Business (mobile –business) is conducting business by exploiting the mobile devices and wireless networking. Mobile business applications include mobile access to existing e-business applications so that mobile users can use these applications from anywhere. It helps to achieve strategic business benefits. Usually, people are seen buying cinema tickets and other tickets using mobile devices.

Managers can use mobile computers in, say, critical presentations to major customers. They can access the latest market share information. They can communicate with the office about possible new offers and call meetings for discussing responds to the new proposals. Many companies send their advertisements on mobiles which helps them to reach large number of users having one – to one correspondence.

Micro-payments are also being done using mobile devices. One can easily get their cable TV connection renewed by making payments using mobiles.

Bills are sent to customers on mobile devices eliminating use of paper work.

Tele-banking is used by banks to execute different transactions.

Transportation industry has really benefited from mobile computing. One can keep track of their consignments. It can also help drivers guide to reach a destination with the shortest possible path available.

In manufacturing industry, due to automation of machines one can control the working of machines from mobile devices. The amount of quantity to be produced can be kept track of using mobile devices.

Thus with the evolution of mobile computing and related devices the business activities have speeded up and is providing huge benefits to companies in growing their interaction with customers.

2. Government

Governments are planning for m-voting, m-interaction and m-administration. In some countries, governments are conducting experiments in this regard. m-administration is concerned with improving the operations and communications with government units.

The government employees can be notified about various policies using mobile devices, resulting into better and faster access to information.

The government-to- citizen interaction is a technique by which interaction with government officials is made easier using mobile devices.

3. In Education field

Mobile computing application is also used in the field of education. Now a days many examination results are displayed on the mobile devices.

Many companies have launched mobile classrooms where students can access education material using mobile devices. Many search engines have added features that help to search books, research papers and other material using mobile devices.

4. In Medical field

For patients whose health is on the line the benefits are even greater. They have increased access to specialized doctors. They don't have to stick around the hospital any longer. This ease in mobility allows them to do their own work while still under the doctor's care.

Safety is another issue that is helped here because the rate of mistakes can significantly be decreased. With Wireless Technologies, their healthcare can be less intrusive, for example in the case of wearable sensors. Patients don't have to show up to the hospital for a blood pressure check. It can be done while they are working through wearing wireless sensors that transmit this information in real-time to their doctors.

5. In banking field

Many banks are offering banking services over Internet, voice and mobile phones through SMS. All updates in transactions are provided to customers on their mobile devices. The people now do not need to go to banks to perform their transactions. People are using plastic money to perform transactions. ATMs wirelessly connected to handheld devices through Bluetooth can be used to operate in remote rural areas.

Credit card authorization is performed on mobile POS terminals equipped with wireless adapters.

6. In Airline industry

Access reservation, flight schedule and ticketing information can be done using mobile devices. Virtual check-in for regular customers, airline baggage and cargo control are done automated machines. Flight arrival and departure information while traveling to the airport is made to people on mobile devices. Also, easier, faster and quicker checking in of baggage by scanning bar-coded information from baggage tickets directly into a database.

Airport security and monitoring is mage easy using mobile devices.

Mobile devices such as PDAs or tablet PCs are used for grading in a simulator training environment.

7. In defense filed

Mobile computing helps in delivering high-performance, energy- efficient solutions in defense field. It provides soldiers access to information anytime, anywhere and improves their situational awareness while using a small, ultra-mobile device. Soldiers can benefit from the reduced size and weight of these devices and have the freedom and flexibility to easily communicate with others.

Also, many devices are now being fitted in fighter planes that helps in better tracking of places.

8. In construction

Recording project data, such as supply delivery records and progress updates, directly at the jobsite is one of the basic functionalities of such systems. More complex systems include a comprehensive suite of hosted internet solutions that facilitate rapid communication and

collaboration throughout the entire project life cycle, from financing and planning through engineering and design, procurement, construction, and facility management.
Document management systems provide construction teams with a project specific extranet, whereby remote team members can communicate and access up-to-date documents. Further, such services alert users affected by changes in real time via e-mail or a pager.

Emergency Services:
Ability to receive information on the move is vital where the emergency services are involved. Information regarding the address, type and other details of an incident can be dispatched quickly, via a Cellular Digital Packet Data (CDPD) system using mobile computers, to one or several appropriate mobile units, which are in the vicinity of the incident.

In courts: Defense counsels can take mobile computers in court. When the opposing counsel references a case which they are not familiar, they can use the computer to get direct, real-time access to on-line legal database services, where they can gather information on the case and related precedents. Therefore, mobile computers allow immediate access to a wealth of information, making people better informed and prepared

Transportation and Shipping
Using mobile computers in conjunction with GPS/GIS and an accompanying vehicle information system (VIS), the operations of an entire transportation fleet can be managed from a central location. The central office knows the location, status, and condition of all vehicles, and operators have two-way communication with the operations center. Using this information, vehicles can be optimally dispatched to maximize efficiency as measured by time; fuel consumption and delivery priority. The mobile computers enable significant performance improvements achieved simultaneously with operational cost reductions.

Farming/ Crop management
Information about weather, crop diseases is provided to farmers on mobile devices. They are also informed about potential places of selling their crops.
Computer systems are used to predict crop diseases and management of the diseases.

REVISION QUESTIONS

- Describe applications areas of Ubiquitous Computing
- Discuss using appropriate examples applications of wireless and mobile technology in the following area:

    i. Education
    ii. Transportation